

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 11

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Direct effect of abiotic change: species near the edge of their range of tolerance (temperature, water) may die out locally, eg sundews if marshes dry, frogs if climate drier; beech (shallow roots) suffers in drought more than oak • Changes in timing: earlier flowering / emergence may separate flowers from pollinators; plant toxins building up earlier may kill caterpillars; warmer winters disturb dormouse hibernation so they use fat reserves and starve • Biotic changes: food supply, predators, disease vectors (mosquitoes / malaria spreading), pollinators and seed dispersers may change • Distribution changes: species may colonise newly suitable areas (triton molluscs around the UK) but slow-growing species (trees) may disappear from old ranges faster than they colonise; suitable areas or biological corridors may not exist; human land use blocks movement • Population fragmentation: rising seas or upward retreat to mountains isolate populations; divided gene pools increase inbreeding; small populations may not be viable; no repopulation from elsewhere • Some species benefit (bats: warmer winters, more insects) but wetter stormier weather reduces feeding time	6
01.2	• All the species it depends on (food plants, pollinators, prey, mycorrhizae) will not move at the same speed • Less mobile species (plants, non-flying animals) cannot cross unsuitable areas without a biological corridor	2
01.3	• The temperature difference between mid-latitude and polar air drives the jet streams; a smaller difference makes the jet stream slower and more meandering (Rossby waves) • Weather systems stall over an area, giving prolonged drought or rainfall / floods; unusually cold or warm air carried north or south	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 90-92; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
02.1	• Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff • Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover	4
02.2	• Tree roots that held the soil together decompose so the soil is bound less strongly • Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely	2
02.3	<i>Any 4 from:</i> • Sedimentation in rivers reduces flow capacity so flooding downstream is more likely • Sedimentation in reservoirs reduces their storage volume • Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants • Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas • Desertification as vegetation cannot re-establish; reduced rainfall • Nutrients (phosphates) carried on soil particles causing eutrophication	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
03.1	• B - Tropospheric ozone (formed by reactions of NO_x in sunlight)	1
03.2	• Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO_x from high-temperature combustion • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO_x dissolve to form nitric acid • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO_x photochemistry	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
04.1	• Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored • Hydrogen has a high energy density so it can be transported / stored for long periods • Released by combustion for heat, industrial processes or steam turbines; or by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water • Can also be used as a vehicle fuel or converted to methane / ammonia	4

Question	Marking guidance	Marks
04.2	<i>Any 3 from:</i> - Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks - Liquefied at very low temperature - energy for refrigeration and compression - Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy - Conversion to ammonia - some energy needed but less than the alternatives	3
04.3	- Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging - A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied	2
04.4	Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
05.1	- Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them - Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade - Disposable products (razors, lighters, pens) that must be replaced - Over-packaging to make products attractive; also increases wastes from manufacture and mineral extraction	4
05.2	<i>Any 3 from:</i> - Properties of the waste: toxicity, flammability, degradability, radioactivity - Population density (collection costs) - Land availability for landfill - Availability of recycling technology / markets - Legislation / regulatory framework (landfill tax) - Public willingness to recycle / environmental awareness - Processing costs / household income	3
05.3	- Advantage: aerobic decomposition produces little methane; nutrients / organic matter returned to soil (circular economy); reduces landfill volume - Disadvantage: needs separation / collection of clean organic waste; odour / pests / pathogens if poorly managed; low-value product; not suitable for mixed waste	2
05.4	Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
06.1	$(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%)	1
06.2	- Although the larger vehicle uses more energy in total, it carries many more people / a larger load - So the energy per passenger (or per tonne of goods) is lower; steel wheels on rails also have low rolling resistance	2

Question	Marking guidance	Marks
06.3	<i>Any 3 from:</i> • Flexibility of routes / destinations and door-to-door convenience • Time / speed and reliability where public transport is infrequent • Low fuel tax / cheap fuel • Carrying goods; comfort; status	3
06.4	• Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey • So more journeys shift from cars to bulk transport / cycling • ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely • Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	
07.1	• Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended • Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges • The particles clump together (flocs) and settle in the clarifier tank	3
07.2	• Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources • Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste	2
07.3	• Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) • Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution)	2
07.4	• Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) • Reverse osmosis / distillation (desalination) - expensive, energy-intensive • Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
08.1	• Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually • BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark) • It gives a single measure directly related to the risk of deoxygenation	3
08.2	• E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is • Pathogens are not always present so their absence would not prove the water is uncontaminated	2

Question	Marking guidance	Marks
08.3	<i>Any 3 from:</i> • Different species have different sensitivities to pollution • Easy to identify • Easy to find / normally present / usually common • Widely distributed	3
08.4	• Advantage: assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone); no expensive equipment needed • Disadvantage: identification needs skill; does not identify which pollutant is present so further chemical tests are required; species may be absent for other reasons	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9.	
09.1	• The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) • Radon travels up cracks in the rock and through foundations into buildings • It becomes concentrated inside enclosed / poorly ventilated buildings; control by ventilating foundations / sumps	3
09.2	• $16 + 1.5 = 17.5\%$	1
09.3	<i>Any 2 from:</i> • Where they live (geology / radon, altitude) • Occupation (radiographer, nuclear worker, aircrew, miner) • Medical history (number of X-rays / CT scans) • Lifestyle / diet (eg shellfish near discharges); smoking	2
09.4	<i>Any 4 from:</i> • X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it; risks can be reduced (shielding, dose limits) • Symptoms of chronic low-dose exposure take a long time to develop • The same symptoms (cancers) have other causes so effects are hard to attribute • Accurate data on human exposure / doses is often unavailable (especially after accidents) • People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Smoke: solid particulates (PM10, PM5, PM1) from incomplete combustion; small particles stay suspended, penetrate lungs; toxic chemicals (acids, phenols, metals) carried on particles; high albedo cools climate; blocks light reducing photosynthesis; damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths); synergism with SO₂ • Controls: Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion; smoke smogs have largely been eliminated in UK cities • Acid rain gases: SO₂ (coal combustion, smelting sulfide ores) oxidised by ozone to SO₃ - sulfuric acid; NO_x (high temperature combustion) - nitric acid; HCl from PVC incineration; reactive gases travel long distances (transboundary pollution) • Effects: pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata); dissolves exoskeletons; leaches Ca and Mg then mobilises toxic Al and Pb; corrodes metals, dissolves limestone; severity depends on bedrock (limestone neutralises) • Controls: fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays; switching from coal to gas / renewables • Evaluation: SO₂ emissions in Europe have fallen dramatically and lake pH is recovering; NO_x harder to control because of growing vehicle numbers; catalytic converters need lead-free fuel and warm engines; FGD is costly and produces wastes / gypsum; controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO₂ vs NO_x	9
10.2	<i>Any 1 from:</i> • Reduces growth rates of trees / crops • Damages leaf cuticles (allowing SO₂ to cause more damage)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25

Question	Marking guidance	Marks
11.2	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25